Brutal Practice Paper B35 - Nutrition in Plants, Fractions & Decimals, Physical & Chemical Changes, Data Handling

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. During a sunlight test a green leaf of area 40 square cm makes 6 milligrams of glucose each hour. A second leaf has only three-fifths of that area. If glucose made is proportional to area, how much does the second leaf make in 2 hours? [Nutrition in Plants]

- **(A)** 3.6 mg
- **(B)** 3 mg
- **(C)** 10 mg
- **(D)** 7.2 mg
- **(E)** 4.8 mg
- **(F)** 12 mg
- **(G)** 14.4 mg

2. Work out 2/3 of 3/4 of 48, and then subtract 5.2 from the result. What is the final value? [Fractions & Decimals]

- **(A)** 29.2
- **(B)** 12
- **(C)** 18.8
- **(D)** 22.8
- **(E)** 19.8
- **(F)** 24

3. When iron rusts, 56 g of iron combines with oxygen to form 80 g of rust. How many grams of oxygen combined with the iron? [Physical & Chemical Changes]

- **(A)** 80 g
- **(B)** 16 g
- **(C)** 8 g
- **(D)** 32 g
- **(E)** 56 g
- **(F)** 136 g
- **(G)** 24 g

4. The midday temperatures for a week were 31, 34, 30, 35, 33, 32 and 29 degrees Celsius. What is the mean temperature? [Data Handling]

- **(A)** 32.5 degrees C
- **(B)** 35 degrees C
- **(C)** 32 degrees C
- **(D)** 30 degrees C
- **(E)** 31 degrees C
- **(F)** 33 degrees C
- **(G)** 224 degrees C

5. A leaf has 12000 stomata, and 5 percent of them stay closed at noon to save water. Of the stomata that are open, one quarter lie on the upper surface. How many open stomata are on the upper surface? [Nutrition in Plants]

- **(A)** 2850 stomata
- **(B)** 3000 stomata
- **(C)** 11400 stomata
- **(D)** 285 stomata
- **(E)** 2280 stomata
- **(F)** 600 stomata

6. A rope is 12.5 m long. You cut off 3/5 of it, then cut 0.75 m from the piece that remains. How long is the final piece? [Fractions & Decimals]

- **(A)** 11.75 m
- **(B)** 3.5 m
- **(C)** 5 m
- **(D)** 5.75 m
- **(E)** 7.5 m
- **(F)** 4.25 m

7. A candle of mass 60 g burns down at 2.5 g per minute. After how many minutes will only 15 g of candle remain? [Physical & Chemical Changes]

- **(A)** 18 minutes
- **(B)** 15 minutes
- **(C)** 22.5 minutes
- **(D)** 45 minutes
- **(E)** 24 minutes
- **(F)** 6 minutes

8. Monthly rainfall over 5 months was 40, 65, 80, 55 and 60 mm. What is the range of this data? [Data Handling]

- **(A)** 20 mm
- **(B)** 300 mm
- **(C)** 45 mm
- **(D)** 40 mm
- **(E)** 60 mm
- **(F)** 80 mm

9. Photosynthesis uses 6 molecules of carbon dioxide and 6 of water to make 1 glucose and 6 oxygen. To make 4 glucose molecules, how many carbon dioxide molecules are needed? [Nutrition in Plants]

- **(A)** 36 molecules
- **(B)** 6 molecules
- **(C)** 30 molecules
- **(D)** 24 molecules
- **(E)** 4 molecules
- **(F)** 10 molecules
- **(G)** 12 molecules

10. Multiply 0.4 by 0.25, then divide the answer by 0.05. What do you get? [Fractions & Decimals]

- **(A)** 0.02
- **(B)** 0.2
- **(C)** 1
- **(D)** 0.5
- **(E)** 20
- **(F)** 2
- **(G)** 5

11. Water freezing into ice is a physical change. 250 ml of water freezes, and ice takes up 9 percent more volume. What is the volume of the ice? [Physical & Chemical Changes]

- **(A)** 259 ml
- **(B)** 22.5 ml
- **(C)** 272.5 ml
- **(D)** 227.5 ml
- **(E)** 250 ml
- **(F)** 225 ml

12. Students grew these numbers of plants: 2, 3, 3, 5, 3, 4, 2, 3. What is the mode of the data? [Data Handling]

- **(A)** 25
- **(B)** 3.125
- **(C)** 2
- **(D)** 8
- **(E)** 5
- **(F)** 4
- **(G)** 3

13. In a variegated leaf, 3/8 of the surface is white with no chlorophyll and the rest is green. If the leaf is 64 square cm, how many square cm can carry out photosynthesis? [Nutrition in Plants]

- **(A)** 30 sq cm
- **(B)** 8 sq cm
- **(C)** 40 sq cm
- **(D)** 16 sq cm
- **(E)** 24 sq cm
- **(F)** 56 sq cm
- **(G)** 48 sq cm

14. Which is larger, 7/8 or 0.82, and by how much? [Fractions & Decimals]

- **(A)** Equal
- **(B)** 7/8 by 0.55
- **(C)** 0.82 by 0.055
- **(D)** 7/8 by 0.075
- **(E)** 7/8 by 0.055
- **(F)** 7/8 by 0.05
- **(G)** 0.82 by 0.02

15. Out of 10 observed changes in a lab, 4 were chemical and the rest physical. What percentage of the changes were physical? [Physical & Chemical Changes]

- **(A)** 4%
- **(B)** 60%
- **(C)** 6%
- **(D)** 10%
- **(E)** 50%
- **(F)** 40%
- **(G)** 66.7%

16. The heights of 6 plants in cm are 12, 18, 9, 15, 21 and 14. What is the median height? [Data Handling]

- **(A)** 21 cm
- **(B)** 14.5 cm
- **(C)** 14.83 cm
- **(D)** 15 cm
- **(E)** 12 cm
- **(F)** 16 cm
- **(G)** 14 cm

17. A legume field of 250 plants fixes 0.4 kg of nitrogen per plant in a season, but 15 percent of it is lost. How many kilograms of nitrogen remain in the soil? [Nutrition in Plants]

- **(A)** 40 kg
- **(B)** 15 kg
- **(C)** 60 kg
- **(D)** 100 kg
- **(E)** 85 kg
- **(F)** 115 kg
- **(G)** 37.5 kg

18. Add 1/4 + 1/6 + 1/3 and give the answer as a decimal. [Fractions & Decimals]

- **(A)** 0.9
- **(B)** 0.75
- **(C)** 0.5
- **(D)** 0.83
- **(E)** 0.66
- **(F)** 1.0

19. Burning magnesium: 24 g of magnesium reacts with 16 g of oxygen to give magnesium oxide. How many grams of oxide form when 48 g of magnesium burns? [Physical & Chemical Changes]

- **(A)** 80 g
- **(B)** 96 g
- **(C)** 64 g
- **(D)** 48 g
- **(E)** 40 g
- **(F)** 56 g
- **(G)** 32 g

20. In a pie chart of 360 degrees, the 'Science' sector covers 25 percent of students' favourite subjects. How many degrees is the Science sector? [Data Handling]

- **(A)** 25 degrees
- **(B)** 120 degrees
- **(C)** 100 degrees
- **(D)** 45 degrees
- **(E)** 90 degrees
- **(F)** 36 degrees

21. Fungi decompose a fallen log, breaking down 0.05 kg of wood each day. How many days will it take them to decompose 0.6 kg of the log? [Nutrition in Plants]

- **(A)** 120 days
- **(B)** 50 days
- **(C)** 30 days
- **(D)** 0.012 days
- **(E)** 6 days
- **(F)** 12 days
- **(G)** 24 days

22. A jug holds 2.4 litres of juice. You pour out 5/8 of it. How many millilitres of juice remain? [Fractions & Decimals]

- **(A)** 1.5 ml
- **(B)** 90 ml
- **(C)** 900 ml
- **(D)** 0.9 ml
- **(E)** 1800 ml
- **(F)** 9000 ml

23. A spring stretches 0.4 cm for every extra 100 g of load, a reversible physical change. How far does it stretch under a 750 g load? [Physical & Chemical Changes]

- **(A)** 30 cm
- **(B)** 0.3 cm
- **(C)** 2.8 cm
- **(D)** 7.5 cm
- **(E)** 2.4 cm
- **(F)** 3 cm

24. The mean of 5 readings is 18. Four of the readings are 15, 20, 17 and 22. What is the fifth reading? [Data Handling]

- **(A)** 22
- **(B)** 74
- **(C)** 14
- **(D)** 20
- **(E)** 16
- **(F)** 18
- **(G)** 90

25. In an iodine starch test, a leaf turns blue-black over 0.75 of its surface. If the leaf area is 36 square cm, what area showed NO starch? [Nutrition in Plants]

- **(A)** 12 sq cm
- **(B)** 24 sq cm
- **(C)** 0.25 sq cm
- **(D)** 27 sq cm
- **(E)** 36 sq cm
- **(F)** 9 sq cm
- **(G)** 18 sq cm

26. Divide 3/4 by 2/3, then multiply the result by 0.5. What is the answer? [Fractions & Decimals]

- **(A)** 1.0
- **(B)** 1.125
- **(C)** 0.625
- **(D)** 2.25
- **(E)** 0.5625
- **(F)** 0.28

27. Curdling of milk is a chemical change. A dairy curdles 4/5 of its 60 litres of milk into curd. How many litres stay as milk? [Physical & Chemical Changes]

- **(A)** 60 litres
- **(B)** 12 litres
- **(C)** 48 litres
- **(D)** 24 litres
- **(E)** 20 litres
- **(F)** 15 litres
- **(G)** 5 litres

28. In a survey, 120 people named a favourite fruit. 35 percent chose mango and 1/4 chose apple. How many more chose mango than apple? [Data Handling]

- **(A)** 72
- **(B)** 25
- **(C)** 12
- **(D)** 18
- **(E)** 42
- **(F)** 7

29. A plant makes 18 mg of glucose per hour during 11 hours of daylight but uses 3 mg per hour for respiration over all 24 hours. What is the net glucose left after one full day? [Nutrition in Plants]

- **(A)** 90 mg
- **(B)** 72 mg
- **(C)** 126 mg
- **(D)** 162 mg
- **(E)** 198 mg
- **(F)** 270 mg
- **(G)** 120 mg

30. A number multiplied by 0.6 gives 4.5. What is the number? [Fractions & Decimals]

- **(A)** 2.7
- **(B)** 3.9
- **(C)** 5.1
- **(D)** 75
- **(E)** 0.75
- **(F)** 7.5
- **(G)** 27

31. Heating copper forms black copper oxide. A 50 g copper strip gains 12.5 percent in mass. What is its new mass? [Physical & Chemical Changes]

- **(A)** 43.75 g
- **(B)** 62.5 g
- **(C)** 50 g
- **(D)** 6.25 g
- **(E)** 56.25 g
- **(F)** 56.5 g

32. A frequency table shows scores: 12 students scored 4, 4 scored 6, and 4 scored 8. What is the mean score? [Data Handling]

- **(A)** 5
- **(B)** 20
- **(C)** 104
- **(D)** 5.2
- **(E)** 4.8
- **(F)** 6

33. Over a cloudy 30-day month, a plant photosynthesised well on only 2/5 of the days. On each good day it stored 12 g of food. How much food did it store that month? [Nutrition in Plants]

- **(A)** 144 g
- **(B)** 132 g
- **(C)** 12 g
- **(D)** 150 g
- **(E)** 240 g
- **(F)** 216 g

34. Write 0.375 as a fraction in its simplest form. [Fractions & Decimals]

- **(A)** 5/8
- **(B)** 37/100
- **(C)** 375/1000
- **(D)** 3/8
- **(E)** 3/4
- **(F)** 3/80
- **(G)** 1/3

35. 5 litres of sea water contain 3.5 percent salt by mass. After all the water evaporates, how many grams of salt remain? (1 litre weighs 1000 g) [Physical & Chemical Changes]

- **(A)** 35 g
- **(B)** 1750 g
- **(C)** 75 g
- **(D)** 17.5 g
- **(E)** 4825 g
- **(F)** 350 g
- **(G)** 175 g

36. A double bar graph shows three clubs. Boys: 12, 15, 9. Girls: 8, 11, 16. How many more children are in the largest club than the smallest? [Data Handling]

- **(A)** 16
- **(B)** 20
- **(C)** 8
- **(D)** 46
- **(E)** 6
- **(F)** 5
- **(G)** 26

37. A leaf loses 0.002 litres of water per hour through its stomata. How many millilitres of water does it lose in 5 hours? [Nutrition in Plants]

- **(A)** 2 ml
- **(B)** 100 ml
- **(C)** 20 ml
- **(D)** 1 ml
- **(E)** 0.01 ml
- **(F)** 5 ml
- **(G)** 10 ml

38. A recipe needs 2 1/2 cups of flour. You decide to make only 1/3 of the recipe. How many cups of flour do you need? [Fractions & Decimals]

- **(A)** 15/2
- **(B)** 1/6
- **(C)** 5/6
- **(D)** 5/2
- **(E)** 5/3
- **(F)** 7/6
- **(G)** 2/3

39. A chemical reaction gives off 0.25 litres of gas for every gram of reactant used. How many litres of gas come from 18 grams of reactant? [Physical & Chemical Changes]

- **(A)** 4.5 litres
- **(B)** 0.45 litres
- **(C)** 18 litres
- **(D)** 2.25 litres
- **(E)** 9 litres
- **(F)** 36 litres
- **(G)** 72 litres

40. The mean weight of 8 boxes is 5.5 kg. One 9 kg box is removed. What is the mean weight of the boxes that remain? [Data Handling]

- **(A)** 44 kg
- **(B)** 35 kg
- **(C)** 5.5 kg
- **(D)** 6 kg
- **(E)** 5 kg
- **(F)** 4.5 kg
- **(G)** 5.2 kg

41. A plant's roots absorb water and dissolved minerals in the volume ratio 9 : 1. If it absorbs 250 ml of liquid in all, how many millilitres are the mineral part? [Nutrition in Plants]

- **(A)** 2.5 ml
- **(B)** 10 ml
- **(C)** 50 ml
- **(D)** 225 ml
- **(E)** 125 ml
- **(F)** 25 ml

42. Evaluate 5.6 divided by 0.08. [Fractions & Decimals]

- **(A)** 70
- **(B)** 56
- **(C)** 4.48
- **(D)** 0.07
- **(E)** 700
- **(F)** 0.7
- **(G)** 7

43. Galvanising coats iron with zinc to prevent rust. A factory coats 240 sheets, but 1/8 of them fail the zinc test. How many sheets pass? [Physical & Chemical Changes]

- **(A)** 30 sheets
- **(B)** 168 sheets
- **(C)** 105 sheets
- **(D)** 32 sheets
- **(E)** 240 sheets
- **(F)** 210 sheets
- **(G)** 200 sheets

44. A bag holds 5 red, 3 blue and 2 green balls. What is the probability of drawing a green ball, as a fraction in simplest form? [Data Handling]

- **(A)** 1/3
- **(B)** 1/2
- **(C)** 2/5
- **(D)** 3/10
- **(E)** 2/8
- **(F)** 1/5
- **(G)** 1/10

45. In a pond survey of 80 organisms, 35 percent are autotrophs that make their own food and the rest are heterotrophs. How many more heterotrophs than autotrophs are there? [Nutrition in Plants]

- **(A)** 28
- **(B)** 52
- **(C)** 12
- **(D)** 24
- **(E)** 45
- **(F)** 80
- **(G)** 30

46. Three-fifths of a class of 40 students are girls. Of those girls, 0.25 wear glasses. How many girls wear glasses? [Fractions & Decimals]

- **(A)** 16
- **(B)** 4
- **(C)** 10
- **(D)** 24
- **(E)** 12
- **(F)** 6
- **(G)** 18

47. 80 g of sugar dissolves fully in hot water; on cooling, 0.65 of it crystallises back out. How many grams stay dissolved? [Physical & Chemical Changes]

- **(A)** 52 g
- **(B)** 15 g
- **(C)** 80 g
- **(D)** 48 g
- **(E)** 65 g
- **(F)** 35 g
- **(G)** 28 g

48. A tally chart records 60 vehicles: 1/3 are cars, 25 percent are bikes, and the rest are trucks. How many trucks are there? [Data Handling]

- **(A)** 45 trucks
- **(B)** 5 trucks
- **(C)** 20 trucks
- **(D)** 40 trucks
- **(E)** 25 trucks
- **(F)** 15 trucks

49. A leaf converts 4/5 of the 60 mg of glucose it makes into stored starch. How many milligrams of glucose are NOT stored? [Nutrition in Plants]

- **(A)** 24 mg
- **(B)** 60 mg
- **(C)** 5 mg
- **(D)** 15 mg
- **(E)** 12 mg
- **(F)** 20 mg
- **(G)** 48 mg

50. Subtract 1 5/6 from 4 1/3. What is the answer as a mixed number? [Fractions & Decimals]

- **(A)** 3 1/6
- **(B)** 3 1/2
- **(C)** 5/6
- **(D)** 3
- **(E)** 2 1/2
- **(F)** 2 2/3
- **(G)** 2 1/6

51. Burning fuel is an irreversible change. A stove burns 0.75 kg of gas per hour. How many hours will a 9 kg cylinder last? [Physical & Chemical Changes]

- **(A)** 6 hours
- **(B)** 6.75 hours
- **(C)** 9 hours
- **(D)** 8 hours
- **(E)** 12 hours
- **(F)** 11.75 hours
- **(G)** 18 hours

52. A cyclist covers 12, 18 and 15 km in three equal one-hour rides. What is the average distance per hour? [Data Handling]

- **(A)** 45 km
- **(B)** 12 km
- **(C)** 22.5 km
- **(D)** 18 km
- **(E)** 14 km
- **(F)** 15 km

53. Three pea roots carry 14, 9 and 22 nitrogen-fixing nodules. What is the average number of nodules per root? [Nutrition in Plants]

- **(A)** 15 nodules
- **(B)** 45 nodules
- **(C)** 14 nodules
- **(D)** 9 nodules
- **(E)** 22 nodules
- **(F)** 12 nodules

54. A driver covers 2/5 of a 150 km trip on day one and 0.35 of the whole trip on day two. How many kilometres are left? [Fractions & Decimals]

- **(A)** 60 km
- **(B)** 112.5 km
- **(C)** 52.5 km
- **(D)** 75 km
- **(E)** 37.5 km
- **(F)** 90 km
- **(G)** 45 km

55. Mixing acid and base is a chemical change that makes salt and water. 30 ml of acid neutralises 45 ml of base. How much base neutralises 50 ml of the same acid? [Physical & Chemical Changes]

- **(A)** 67.5 ml
- **(B)** 90 ml
- **(C)** 75 ml
- **(D)** 65 ml
- **(E)** 60 ml
- **(F)** 50 ml

56. A bar graph shows 200 trees: 30 percent oak, 0.45 pine, and the rest teak. How many teak trees are there? [Data Handling]

- **(A)** 50 teak
- **(B)** 60 teak
- **(C)** 75 teak
- **(D)** 150 teak
- **(E)** 90 teak
- **(F)** 25 teak
- **(G)** 100 teak

57. Air is about 0.04 percent carbon dioxide by volume. A greenhouse holds 50000 litres of air. How many litres of carbon dioxide does it contain? [Nutrition in Plants]

- **(A)** 4 litres
- **(B)** 2 litres
- **(C)** 2000 litres
- **(D)** 25 litres
- **(E)** 0.2 litres
- **(F)** 200 litres
- **(G)** 20 litres

58. Add 7/10 + 3/100 + 9/1000 and write the answer as a decimal. [Fractions & Decimals]

- **(A)** 0.739
- **(B)** 0.937
- **(C)** 0.0739
- **(D)** 0.937
- **(E)** 0.793
- **(F)** 0.73

59. Ripening of fruit is a chemical change. In a basket of 50 fruits, 0.4 are ripe today and 30 percent of the rest ripen tomorrow. How many ripen tomorrow? [Physical & Chemical Changes]

- **(A)** 20 fruits
- **(B)** 21 fruits
- **(C)** 30 fruits
- **(D)** 12 fruits
- **(E)** 6 fruits
- **(F)** 15 fruits
- **(G)** 9 fruits

60. The mean of four numbers is 25. Three of them are 20, 30 and 28. By how much is the fourth number below the mean? [Data Handling]

- **(A)** 3 above the mean
- **(B)** 5 below the mean
- **(C)** 3 below the mean
- **(D)** 22
- **(E)** 22 below the mean
- **(F)** 78

61. A Venus flytrap digests an insect in 9 days and catches a new insect every 12 days. What fraction of each 12-day cycle is spent digesting, in simplest form? [Nutrition in Plants]

- **(A)** 3/12
- **(B)** 1/4
- **(C)** 2/3
- **(D)** 3/4
- **(E)** 9/12
- **(F)** 12/9
- **(G)** 4/3

62. A water tank is 3/4 full and holds 90 litres at that level. How many litres does the full tank hold? [Fractions & Decimals]

- **(A)** 112.5 litres
- **(B)** 135 litres
- **(C)** 90 litres
- **(D)** 67.5 litres
- **(E)** 30 litres
- **(F)** 120 litres
- **(G)** 360 litres

63. Melting ice into water is a physical change with no mass change. 1.5 kg of ice melts completely. How many grams of water form? [Physical & Chemical Changes]

- **(A)** 1500 g
- **(B)** 1650 g
- **(C)** 1350 g
- **(D)** 150 g
- **(E)** 15 g
- **(F)** 1.5 g
- **(G)** 15000 g

64. A line graph shows a plant's height as 4, 7, 11 and 16 cm over four weeks. In which week was the growth greatest, and by how much? [Data Handling]

- **(A)** Week 4, by 5 cm
- **(B)** Week 2, by 3 cm
- **(C)** Equal each week
- **(D)** Week 4, by 11 cm
- **(E)** Week 1, by 4 cm
- **(F)** Week 3, by 4 cm
- **(G)** Week 4, by 16 cm

65. A plant short of magnesium makes only 65 percent of its normal 240 mg of chlorophyll. By how many milligrams is it short of the normal amount? [Nutrition in Plants]

- **(A)** 24 mg
- **(B)** 75 mg
- **(C)** 168 mg
- **(D)** 35 mg
- **(E)** 96 mg
- **(F)** 156 mg
- **(G)** 84 mg

66. Divide 9 by 2/3, then subtract 0.5 from the result. What is the answer? [Fractions & Decimals]

- **(A)** 13
- **(B)** 12.5
- **(C)** 6
- **(D)** 5.5
- **(E)** 14
- **(F)** 13.5

67. Digestion is a chemical change. A meal carrying 600 kJ of energy is 65 percent absorbed; the rest passes out. How many kilojoules are absorbed? [Physical & Chemical Changes]

- **(A)** 195 kJ
- **(B)** 210 kJ
- **(C)** 65 kJ
- **(D)** 360 kJ
- **(E)** 420 kJ
- **(F)** 390 kJ
- **(G)** 35 kJ

68. A pictograph uses 1 symbol to stand for 8 books. A shelf is shown by 5 1/2 symbols. How many books are on the shelf? [Data Handling]

- **(A)** 13.5 books
- **(B)** 88 books
- **(C)** 44 books
- **(D)** 40 books
- **(E)** 48 books
- **(F)** 5.5 books
- **(G)** 36 books

69. Leaf P makes 8 mg of glucose in 30 minutes; leaf Q makes 15 mg in 60 minutes. Which leaf has the faster rate, and by how much per hour? [Nutrition in Plants]

- **(A)** P by 1 mg per hour
- **(B)** P by 7 mg per hour
- **(C)** Q by 1 mg per hour
- **(D)** P by 2 mg per hour
- **(E)** Equal rates
- **(F)** Q by 7 mg per hour
- **(G)** P by 8 mg per hour

70. A wire 6.5 m long is cut into pieces each 0.8 m long. How many full pieces are cut, and what length is left over? [Fractions & Decimals]

- **(A)** 6 pieces, 1.7 m left
- **(B)** 7 pieces, 0.9 m left
- **(C)** 81 pieces, none left
- **(D)** 8 pieces, 0.1 m left
- **(E)** 8 pieces, 0.2 m left
- **(F)** 8 pieces, 0.5 m left

71. Folding paper is a physical change. A sheet is folded in half 4 times in a row. Into how many layers is it divided? [Physical & Chemical Changes]

- **(A)** 12 layers
- **(B)** 8 layers
- **(C)** 16 layers
- **(D)** 4 layers
- **(E)** 32 layers
- **(F)** 6 layers

72. The mean of 6 numbers is 14. Adding a 7th number raises the mean to 15. What is the 7th number? [Data Handling]

- **(A)** 1
- **(B)** 84
- **(C)** 105
- **(D)** 16
- **(E)** 14
- **(F)** 21
- **(G)** 15

73. A plant stores 2000 units of energy. A deer that eats it keeps only 1/10, and a tiger that eats the deer keeps 1/10 of that. How many units reach the tiger? [Nutrition in Plants]

- **(A)** 10 units
- **(B)** 18 units
- **(C)** 40 units
- **(D)** 20 units
- **(E)** 100 units
- **(F)** 2 units
- **(G)** 200 units

74. Multiply 1 2/5 by 2.5. What is the answer? [Fractions & Decimals]

- **(A)** 3.0
- **(B)** 4.5
- **(C)** 3.5
- **(D)** 2.9
- **(E)** 7
- **(F)** 5.6
- **(G)** 2.5

75. Souring of milk is a chemical change. Milk turns sour in 6 hours at 30 degrees Celsius but twice as fast at 40 degrees. How many hours does it take at 40 degrees? [Physical & Chemical Changes]

- **(A)** 9 hours
- **(B)** 8 hours
- **(C)** 6 hours
- **(D)** 4 hours
- **(E)** 1.5 hours
- **(F)** 3 hours

76. A spinner has 8 equal parts: 3 red and 5 yellow. What is the probability of NOT landing on red, as a fraction? [Data Handling]

- **(A)** 1/2
- **(B)** 5/8
- **(C)** 5/3
- **(D)** 3/5
- **(E)** 1/8
- **(F)** 8/5

77. A tree loses 1.25 litres of water per hour during 9 daytime hours and 0.4 litres per hour during 15 night hours. How many litres does it lose in a full day? [Nutrition in Plants]

- **(A)** 15 litres
- **(B)** 39.6 litres
- **(C)** 17.25 litres
- **(D)** 17.65 litres
- **(E)** 6 litres
- **(F)** 18 litres
- **(G)** 11.25 litres

78. Among 0.6, 5/8, 0.58 and 3/5, which value is the largest? [Fractions & Decimals]

- **(A)** All equal
- **(B)** 0.58
- **(C)** 0.6 and 3/5
- **(D)** 5/8
- **(E)** 3/5
- **(F)** 0.6
- **(G)** 0.58

79. Combustion needs oxygen, which is 21 percent of air. A sealed room holds 4000 litres of air. How many litres of oxygen are available to a flame? [Physical & Chemical Changes]

- **(A)** 800 litres
- **(B)** 840 litres
- **(C)** 84 litres
- **(D)** 79 litres
- **(E)** 21 litres
- **(F)** 8400 litres
- **(G)** 420 litres

80. In a class of 50 students, 12 scored below 40, 20 scored between 40 and 60, and the rest scored above 60. What percentage scored above 60? [Data Handling]

- **(A)** 36%
- **(B)** 18%
- **(C)** 64%
- **(D)** 32%
- **(E)** 40%
- **(F)** 30%
- **(G)** 24%

81. One leaf cell holds about 50 chloroplasts, and a small leaf has 2000 such cells. How many chloroplasts does the leaf hold in total? [Nutrition in Plants]

- **(A)** 10000
- **(B)** 50000
- **(C)** 1000000
- **(D)** 250000
- **(E)** 40000
- **(F)** 100000
- **(G)** 2050

82. A shopkeeper sells 3/4 kg, 0.6 kg and 5/8 kg of sugar to three customers. What is the total weight sold? [Fractions & Decimals]

- **(A)** 1.975 kg
- **(B)** 1.9 kg
- **(C)** 1.75 kg
- **(D)** 2.025 kg
- **(E)** 2.0 kg
- **(F)** 1.875 kg

83. Boiling and condensing water is a reversible physical change. 2 litres of water boil away in 25 minutes. What is the boiling rate in millilitres per minute? [Physical & Chemical Changes]

- **(A)** 800 ml per minute
- **(B)** 80 ml per minute
- **(C)** 100 ml per minute
- **(D)** 50 ml per minute
- **(E)** 40 ml per minute
- **(F)** 12.5 ml per minute
- **(G)** 8 ml per minute

84. The mean rainfall of 4 cities is 80 mm. Three cities had 70, 95 and 60 mm. The fourth city's rainfall is what fraction of the total, in simplest form? [Data Handling]

- **(A)** 19/64
- **(B)** 3/10
- **(C)** 1/4
- **(D)** 95/320
- **(E)** 19/60
- **(F)** 1/3
- **(G)** 95/225

85. Fresh spinach is 91 percent water. A bunch weighs 350 g fresh. After it is fully dried, only the non-water part remains. What is the dry mass? [Nutrition in Plants]

- **(A)** 9 g
- **(B)** 41.5 g
- **(C)** 315 g
- **(D)** 31.5 g
- **(E)** 63 g
- **(F)** 318.5 g
- **(G)** 35 g

86. Find 2/3 of 7/8 of 0.96. What is the result? [Fractions & Decimals]

- **(A)** 0.72
- **(B)** 0.6
- **(C)** 0.84
- **(D)** 0.42
- **(E)** 0.64
- **(F)** 0.56

87. To make strong mortar, a builder mixes cement and sand in the mass ratio 1 : 4. To make 60 kg of mortar, how many kilograms of cement are needed? [Physical & Chemical Changes]

- **(A)** 48 kg
- **(B)** 6 kg
- **(C)** 12 kg
- **(D)** 4 kg
- **(E)** 15 kg
- **(F)** 30 kg
- **(G)** 20 kg

88. Daily steps were Mon 6000, Tue 8000, Wed 5000, Thu 9000, Fri 7000. What is the mean number of steps per day? [Data Handling]

- **(A)** 9000 steps
- **(B)** 7500 steps
- **(C)** 5000 steps
- **(D)** 8000 steps
- **(E)** 6000 steps
- **(F)** 7000 steps

89. In one hour a tree absorbs 48 g of carbon dioxide while a small shrub absorbs 18 g. What is the ratio of the tree's to the shrub's absorption, in simplest form? [Nutrition in Plants]

- **(A)** 24 : 9
- **(B)** 16 : 6
- **(C)** 2 : 1
- **(D)** 5 : 2
- **(E)** 48 : 18
- **(F)** 8 : 3

90. A pizza is cut into 8 equal slices. You eat 3 slices and your friend eats 0.25 of the whole pizza. What fraction of the pizza is left? [Fractions & Decimals]

- **(A)** 3/4
- **(B)** 2/8
- **(C)** 1/8
- **(D)** 1/2
- **(E)** 1/4
- **(F)** 5/8
- **(G)** 3/8

91. Electrolysis splits 36 g of water into hydrogen and oxygen in the mass ratio 1 : 8. How many grams of oxygen are produced? [Physical & Chemical Changes]

- **(A)** 28 g
- **(B)** 18 g
- **(C)** 4 g
- **(D)** 8 g
- **(E)** 36 g
- **(F)** 9 g
- **(G)** 32 g

92. In a data set of 9 values arranged in increasing order, which value is the median? [Data Handling]

- **(A)** The 4th value
- **(B)** The 6th value
- **(C)** The 9th value
- **(D)** The average of all values
- **(E)** The 1st value
- **(F)** The 5th value

93. A leaf makes 0.6 g of glucose in sunlight, loses 0.25 g of it to respiration, then turns 0.2 g of what remains into starch. How much glucose stays as free sugar? [Nutrition in Plants]

- **(A)** 0.35 g
- **(B)** 0.2 g
- **(C)** 0.4 g
- **(D)** 0.05 g
- **(E)** 0.45 g
- **(F)** 0.15 g
- **(G)** 0.55 g

94. Write 2.625 as a mixed number in its simplest form. [Fractions & Decimals]

- **(A)** 2 5/8
- **(B)** 2 625/1000
- **(C)** 2 13/16
- **(D)** 2 3/4
- **(E)** 2 2/3
- **(F)** 2 1/4
- **(G)** 2 1/2

95. Polishing a silver spoon removes 0.6 percent of its 40 g tarnish-coated mass. How many grams are removed? [Physical & Chemical Changes]

- **(A)** 0.06 g
- **(B)** 0.6 g
- **(C)** 0.4 g
- **(D)** 0.024 g
- **(E)** 2.4 g
- **(F)** 24 g
- **(G)** 0.24 g

96. A class of 25 students has a mean test score of 60. The 20 boys averaged 55. The other 5 are girls. What was the girls' average score? [Data Handling]

- **(A)** 60
- **(B)** 400
- **(C)** 65
- **(D)** 70
- **(E)** 75
- **(F)** 80

97. A flower of an insect-eating plant lures 3 insects on day one, and each day after it lures 3 more than the day before. How many insects does it lure on the 6th day? [Nutrition in Plants]

- **(A)** 15 insects
- **(B)** 21 insects
- **(C)** 24 insects
- **(D)** 33 insects
- **(E)** 18 insects
- **(F)** 9 insects
- **(G)** 6 insects

98. Evaluate 0.45 + 1/5 - 0.3. What is the answer? [Fractions & Decimals]

- **(A)** 0.3
- **(B)** 0.15
- **(C)** 0.65
- **(D)** 0.95
- **(E)** 0.35
- **(F)** 0.55

99. A 100 g iron nail rusts, gaining 8 percent in mass; then 25 percent of the rust formed flakes off. What is the nail's final mass? [Physical & Chemical Changes]

- **(A)** 108 g
- **(B)** 100 g
- **(C)** 106 g
- **(D)** 104 g
- **(E)** 110 g
- **(F)** 102 g
- **(G)** 81 g

100. In a daily-activity pie chart, the 'sleep' sector measures 120 degrees. Out of a 24-hour day, how many hours does this represent? [Data Handling]

- **(A)** 120 hours
- **(B)** 4 hours
- **(C)** 8 hours
- **(D)** 9 hours
- **(E)** 10 hours
- **(F)** 12 hours